

Arithmetic Sequences and Series (7.2)

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: -3

_____ Find the common difference of $3, 7, 11, 15, \dots$

ANSWER: 7

_____ Find S_5 for $a_1 = 3$ and $a_5 = 19$.

ANSWER: $3n - 1$

_____ Find a_{10} for $a_1 = 5$, $d = 2$.

ANSWER: 59

_____ Find the sum of the first 10 terms of $1, 3, 5, \dots$

ANSWER: 4 # _____ Write an explicit formula for the arithmetic sequence with $a_1 = 2$, $d = 3$.	ANSWER: 3 # _____ Find the 20th term of $2, 5, 8, \dots$
ANSWER: 5050 # _____ If $a_4 = 24$ and $a_{13} = 87$, find the common difference d.	ANSWER: 100 # _____ If $a_1 = 10$ and $a_6 = -5$, find the common difference d.
ANSWER: 23 # _____ Find $1 + 2 + 3 + \dots + 100$.	ANSWER: 55 # _____ Insert three arithmetic means between 5 and 17; give the common difference.